

MongoDB Lab – Day 5 (Indexing & Sorting)

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CSE 206

Schema { emp_id, name, department, designation, city, salary, experience }

Initial Data:

Emp ID	Name	Department	Designation	City	Salary	Experience
101	Amit	IT	Developer	Kolkata	60000	3
102	Neha	HR	HR Executive	Delhi	45000	2
103	Ravi	IT	Tester	Mumbai	50000	4
104	Pooja	Finance	Accountant	Chennai	55000	5
105	Karan	IT	Developer	Kolkata	65000	6
106	Sneha	HR	Manager	Delhi	70000	7
107	Arjun	Finance	Analyst	Mumbai	48000	2
108	Meera	IT	Developer	Bangalore	75000	8
109	Rohit	IT	Team Lead	Pune	90000	10
110	Anjali	HR	Recruiter	Kolkata	42000	1

Inserting Documents:

```
test> use college_CSE_206
switched to db college_CSE_206
college_CSE_206> db.student_CSE_206.insertMany([
... {emp_id:101,name:"Amit",department:"IT",designation:"Developer",city:"Kolkata",salary:60000,experience:3},
... {emp_id:102,name:"Neha",department:"HR",designation:"HR Executive",city:"Delhi",salary:45000,experience:2},
... {emp_id:103,name:"Ravi",department:"IT",designation:"Tester",city:"Mumbai",salary:50000,experience:4},
... {emp_id:104,name:"Pooja",department:"Finance",designation:"Accountant",city:"Chennai",salary:55000,experience:5},
... {emp_id:105,name:"Karan",department:"IT",designation:"Developer",city:"Kolkata",salary:65000,experience:6},
... {emp_id:106,name:"Sneha",department:"HR",designation:"Manager",city:"Delhi",salary:70000,experience:7},
... {emp_id:107,name:"Arjun",department:"Finance",designation:"Analyst",city:"Mumbai",salary:48000,experience:2},
... {emp_id:108,name:"Meera",department:"IT",designation:"Developer",city:"Bangalore",salary:75000,experience:8},
... {emp_id:109,name:"Rohit",department:"IT",designation:"Team Lead",city:"Pune",salary:90000,experience:10},
... {emp_id:110,name:"Anjali",department:"HR",designation:"Recruiter",city:"Kolkata",salary:42000,experience:1}
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699aaab1b1839f13191e2621'),
    '1': ObjectId('699aaab1b1839f13191e2622'),
    '2': ObjectId('699aaab1b1839f13191e2623'),
    '3': ObjectId('699aaab1b1839f13191e2624'),
    '4': ObjectId('699aaab1b1839f13191e2625'),
    '5': ObjectId('699aaab1b1839f13191e2626'),
    '6': ObjectId('699aaab1b1839f13191e2627'),
    '7': ObjectId('699aaab1b1839f13191e2628'),
    '8': ObjectId('699aaab1b1839f13191e2629'),
    '9': ObjectId('699aaab1b1839f13191e262a')
  }
}
```

Lab Questions:

Q1. Display all employee records.

Query: db.student_CSE_206.find()

Output:-

```
{
  _id: ObjectId('699aaab1b1839f13191e2621'),
  emp_id: 101,
  name: 'Amit',
  department: 'IT',
  designation: 'Developer',
  city: 'Kolkata',
  salary: 60000,
  experience: 3
},
{
  _id: ObjectId('699aaab1b1839f13191e2622'),
  emp_id: 102,
  name: 'Neha',
  department: 'HR',
  designation: 'HR Executive',
  city: 'Delhi',
  salary: 45000,
  experience: 2
},
{
  _id: ObjectId('699aaab1b1839f13191e2623'),
  emp_id: 103,
  name: 'Ravi',
  department: 'IT',
  designation: 'Tester',
  city: 'Mumbai',
  salary: 50000,
  experience: 4
},
{
  _id: ObjectId('699aaab1b1839f13191e2624'),
  emp_id: 104,
  name: 'Pooja',
  department: 'Finance',
  designation: 'Accountant',
  city: 'Chennai',
  salary: 55000,
  experience: 5
},
}
```

```
{
  _id: ObjectId('699aaab1b1839f13191e2625'),
  emp_id: 105,
  name: 'Karan',
  department: 'IT',
  designation: 'Developer',
  city: 'Kolkata',
  salary: 65000,
  experience: 6
},
{
  _id: ObjectId('699aaab1b1839f13191e2626'),
  emp_id: 106,
  name: 'Sneha',
  department: 'HR',
  designation: 'Manager',
  city: 'Delhi',
  salary: 70000,
  experience: 7
},
{
  _id: ObjectId('699aaab1b1839f13191e2627'),
  emp_id: 107,
  name: 'Arjun',
  department: 'Finance',
  designation: 'Analyst',
  city: 'Mumbai',
  salary: 48000,
  experience: 2
},
{
  _id: ObjectId('699aaab1b1839f13191e2628'),
  emp_id: 108,
  name: 'Meera',
  department: 'IT',
  designation: 'Developer',
  city: 'Bangalore',
  salary: 75000,
  experience: 8
},
}
```

```

{
  _id: ObjectId('699aaab1b1839f13191e2629'),
  emp_id: 109,
  name: 'Rohit',
  department: 'IT',
  designation: 'Team Lead',
  city: 'Pune',
  salary: 90000,
  experience: 10
},
{
  _id: ObjectId('699aaab1b1839f13191e262a'),
  emp_id: 110,
  name: 'Anjali',
  department: 'HR',
  designation: 'Recruiter',
  city: 'Kolkata',
  salary: 42000,
  experience: 1
}

```

Q2. Create an index on emp_id field.

Query: db.student_CSE_206.createIndex({emp_id:1})

Output: emp_id_1

Q3. Create an index on department field.

Query: db.student_CSE_206.createIndex({department:1})

Output: department_1

Q4. Create a compound index on department and salary.

Query: db.student_CSE_206.createIndex({department:1,salary:1})

Output: department_1_salary_1

Q5. Find employees working in IT department.

Query: db.student_CSE_206.find({department:"IT"})

Output:

```
[
  {
    _id: ObjectId('699aaab1b1839f13191e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('699aaab1b1839f13191e2623'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 50000,
    experience: 4
  },
  {
    _id: ObjectId('699aaab1b1839f13191e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('699aaab1b1839f13191e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
]
```

```
{
  _id: ObjectId('699aaab1b1839f13191e2629'),
  emp_id: 109,
  name: 'Rohit',
  department: 'IT',
  designation: 'Team Lead',
  city: 'Pune',
  salary: 90000,
  experience: 10
}
```

Q6. Display employees whose salary is greater than 60,000.

Query: db.student_CSE_206.find({salary:{\$gt:60000}})

Output:

```
[
  {
    _id: ObjectId('699aaab1b1839f13191e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('699aaab1b1839f13191e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('699aaab1b1839f13191e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('699aaab1b1839f13191e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  }
]
```

Q7. Sort employees by salary in descending order.

Query: db.student_CSE_206.find().sort({salary:-1})

Output:

```
[
  {
    _id: ObjectId('699aaab1b1839f13191e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  },
  {
    _id: ObjectId('699aaab1b1839f13191e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('699aaab1b1839f13191e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('699aaab1b1839f13191e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  }
]
```

```
{
  _id: ObjectId('699aaab1b1839f13191e2621'),
  emp_id: 101,
  name: 'Amit',
  department: 'IT',
  designation: 'Developer',
  city: 'Kolkata',
  salary: 60000,
  experience: 3
},
{
  _id: ObjectId('699aaab1b1839f13191e2624'),
  emp_id: 104,
  name: 'Pooja',
  department: 'Finance',
  designation: 'Accountant',
  city: 'Chennai',
  salary: 55000,
  experience: 5
},
{
  _id: ObjectId('699aaab1b1839f13191e2623'),
  emp_id: 103,
  name: 'Ravi',
  department: 'IT',
  designation: 'Tester',
  city: 'Mumbai',
  salary: 50000,
  experience: 4
},
{
  _id: ObjectId('699aaab1b1839f13191e2627'),
  emp_id: 107,
  name: 'Arjun',
  department: 'Finance',
  designation: 'Analyst',
  city: 'Mumbai',
  salary: 48000,
  experience: 2
}
]
```

```

    {
      _id: ObjectId('699aaab1b1839f13191e2622'),
      emp_id: 102,
      name: 'Neha',
      department: 'HR',
      designation: 'HR Executive',
      city: 'Delhi',
      salary: 45000,
      experience: 2
    },
    {
      _id: ObjectId('699aaab1b1839f13191e262a'),
      emp_id: 110,
      name: 'Anjali',
      department: 'HR',
      designation: 'Recruiter',
      city: 'Kolkata',
      salary: 42000,
      experience: 1
    }
  ]

```

Q8. Display top 3 highest paid employees.

Query: `db.student_CSE_206.find().sort({salary:-1}).limit(3)`

Output:

```

[
  {
    _id: ObjectId('699aaab1b1839f13191e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  },
  {
    _id: ObjectId('699aaab1b1839f13191e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('699aaab1b1839f13191e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  }
]

```

Q9. Display employees having experience more than 5 years.

Query: db.student_CSE_206.find({experience:{\$gt:5}})

Output:

```
{
  _id: ObjectId('699aaab1b1839f13191e2625'),
  emp_id: 105,
  name: 'Karan',
  department: 'IT',
  designation: 'Developer',
  city: 'Kolkata',
  salary: 65000,
  experience: 6
},
{
  _id: ObjectId('699aaab1b1839f13191e2626'),
  emp_id: 106,
  name: 'Sneha',
  department: 'HR',
  designation: 'Manager',
  city: 'Delhi',
  salary: 70000,
  experience: 7
},
{
  _id: ObjectId('699aaab1b1839f13191e2628'),
  emp_id: 108,
  name: 'Meera',
  department: 'IT',
  designation: 'Developer',
  city: 'Bangalore',
  salary: 75000,
  experience: 8
},
{
  _id: ObjectId('699aaab1b1839f13191e2629'),
  emp_id: 109,
  name: 'Rohit',
  department: 'IT',
  designation: 'Team Lead',
  city: 'Pune',
  salary: 90000,
  experience: 10
}
```


Q10. Analyze query performance using explain().

Query: db.student_CSE_206.find({department:"IT"}).explain()

Output:

```
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'college_CSE_206.student_CSE_206',
    parsedQuery: { department: { '$eq': 'IT' } },
    indexFilterSet: false,
    queryHash: 'EB9C5E66',
    planCacheShapeHash: 'EB9C5E66',
    planCacheKey: '25CB5D62',
    optimizationTimeMillis: 2,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      inputStage: {
        stage: 'IXSCAN',
        keyPattern: { department: 1 },
        indexName: 'department_1',
        isMultiKey: false,
        multiKeyPaths: { department: [] },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: { department: [ '["IT", "IT"]' ] }
      }
    }
  },
}
```

```

rejectedPlans: [
  {
    isCached: false,
    stage: 'FETCH',
    inputStage: {
      stage: 'IXSCAN',
      keyPattern: { department: 1, salary: 1 },
      indexName: 'department_1_salary_1',
      isMultiKey: false,
      multiKeyPaths: { department: [], salary: [] },
      isUnique: false,
      isSparse: false,
      isPartial: false,
      indexVersion: 2,
      direction: 'forward',
      indexBounds: {
        department: [ '['IT', "IT"]' ],
        salary: [ '[MinKey, MaxKey]' ]
      }
    }
  }
],
queryShapeHash: 'D2109357F4AFD86D678BC6AB696E1E1B0AA39F22EA3420EEDC214B7293ABF017',
command: {
  find: 'student_CSE_206',
  filter: { department: 'IT' },
  '$db': 'college_CSE_206'
},
serverInfo: {
  host: 'LAPTOP-JSNHNSUK',
  port: 27017,
  version: '8.2.3',
  gitVersion: '36f41c9c30a2f13f834d033ba03c3463c891fb01'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,

```

```

serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}

```

```
_id: ObjectId('699aaab1b1839f13191e2621')
emp_id: 101
name: "Amit"
department: "IT"
designation: "Developer"
city: "Kolkata"
salary: 60000
experience: 3
```

```
_id: ObjectId('699aaab1b1839f13191e2622')
emp_id: 102
name: "Neha"
department: "HR"
designation: "HR Executive"
city: "Delhi"
salary: 45000
experience: 2
```

```
_id: ObjectId('699aaab1b1839f13191e2623')
emp_id: 103
name: "Ravi"
department: "IT"
designation: "Tester"
city: "Mumbai"
salary: 50000
experience: 4
```

```
_id: ObjectId('699aaab1b1839f13191e2624')
emp_id: 104
name: "Pooja"
department: "Finance"
designation: "Accountant"
city: "Chennai"
salary: 55000
experience: 5
```

```
_id: ObjectId('699aaab1b1839f13191e2625')
emp_id: 105
name: "Karan"
department: "IT"
designation: "Developer"
city: "Kolkata"
salary: 65000
experience: 6
```

```
_id: ObjectId('699aaab1b1839f13191e2626')
emp_id: 106
name: "Sneha"
department: "HR"
designation: "Manager"
city: "Delhi"
salary: 70000
experience: 7
```

```
_id: ObjectId('699aaab1b1839f13191e2627')
emp_id: 107
name: "Arjun"
department: "Finance"
designation: "Analyst"
city: "Mumbai"
salary: 48000
experience: 2
```

```
_id: ObjectId('699aaab1b1839f13191e2628')
emp_id: 108
name: "Meera"
department: "IT"
designation: "Developer"
city: "Bangalore"
salary: 75000
experience: 8
```

```
_id: ObjectId('699aaab1b1839f13191e2629')
emp_id: 109
name: "Rohit"
department: "IT"
designation: "Team Lead"
city: "Pune"
salary: 90000
experience: 10
```

```
_id: ObjectId('699aaab1b1839f13191e262a')
emp_id: 110
name: "Anjali"
department: "HR"
designation: "Recruiter"
city: "Kolkata"
salary: 42000
experience: 1
```
